



High-Dimensional Statistics

Fall 2025

Prof. Mohammad Hossein Rohban
Sharif University of Technology

Assignment 2

Instructions

- **Due.** December 4, 2025
- **Scoring.** The assignment contains **145 points** in total, but **100 points** are sufficient to receive full credit. Extra points will not be carried over to other assignments.
- **Late policy.** A maximum of three assignments may be submitted late, each with a delay of up to 3 days. Submissions beyond this limit or after the allowed delay will not be accepted.
- **Collaboration.** If you discuss concepts with classmates, list the collaborators.
- State assumptions, cite any less well-known lemmas or results you use, and keep proofs concise.

Problem 1. Properties of ℓ_q -balls

[15 points]

For a given $q \in (0, 1]$, recall the (strong) ℓ_q -ball

$$B_q(R_q) := \left\{ \theta \in \mathbb{R}^d \mid \sum_{j=1}^d |\theta_j|^q \leq R_q \right\}.$$

The weak ℓ_q -ball with parameters (C, α) is defined as

$$B_{w(\alpha)}(C) := \left\{ \theta \in \mathbb{R}^d \mid |\theta|_{(j)} \leq C j^{-\alpha} \text{ for } j = 1, \dots, d \right\},$$

where $|\theta|_{(j)}$ denotes the order statistics of θ in absolute value, ordered from largest to smallest (so that $|\theta|_{(1)} = \max_j |\theta_j|$ and $|\theta|_{(d)} = \min_j |\theta_j|$).

- (7 points) Show that the set $B_q(R_q)$ is star-shaped around the origin. (A set $C \subseteq \mathbb{R}^d$ is star-shaped around the origin if $\theta \in C \Rightarrow t\theta \in C$ for all $t \in [0, 1]$.)
- (8 points) For any $\alpha > 1/q$, show that there is a radius R_q depending on (C, α) such that

$$B_{w(\alpha)}(C) \subseteq B_q(R_q).$$

This inclusion underlies the terminology “strong” and “weak”, respectively.

Problem 2. Welch Bound

[15 points]

Let $A \in \mathbb{C}^{m \times n}$ be a matrix with normalized columns. Prove that

$$\mu(A) \geq \sqrt{\frac{n-m}{m(n-1)}},$$

where $\mu(A)$ is the pairwise incoherence parameter for matrix A .

Hint. Let $a_1, \dots, a_n$ be the normalized columns of A and let $G = A^*A$ be the $n \times n$ Gram matrix with entries $G_{ij} = \langle a_i, a_j \rangle$. Use that:

- $\text{tr}(G) = n$ and $\text{rank}(G) \leq m$,
- for any positive semidefinite matrix G , $(\text{tr } G)^2 \leq \text{rank}(G) \text{tr}(G^2)$,
- $\text{tr}(G^2) = \sum_{i,j} |\langle a_i, a_j \rangle|^2$, and each off-diagonal term is bounded by $\mu(A)^2$.

Problem 3. Mutual Coherence**[40 points]**

Recall that for an arbitrary pair of orthonormal bases

$$\Psi = [\psi_1, \dots, \psi_n] \in \mathbb{R}^{n \times n}, \quad \Phi = [\phi_1, \dots, \phi_n] \in \mathbb{R}^{n \times n},$$

the mutual coherence of these two bases is defined as

$$\mu(\Psi, \Phi) = \max_{1 \leq i, j \leq n} |\psi_i^\top \phi_j|.$$

(a) (10 points) Show that

$$\frac{1}{\sqrt{n}} \leq \mu(\Psi, \Phi) \leq 1.$$

(b) (15 points) Let $\Psi = I$, and suppose that $\Phi = [\phi_{ij}]_{1 \leq i, j \leq n}$ is a Gaussian random matrix such that the entries $\phi_{ij} \sim \mathcal{N}(0, 1/n)$ are i.i.d. random variables. Provide an upper estimate on $\mu(\Psi, \Phi)$ as a function $f(n)$ such that

$$\mathbb{P}\{\mu(\Psi, \Phi) > f(n)\} \rightarrow 0 \quad \text{as } n \rightarrow \infty.$$

Hint: You may use the crude approximation $\mathbb{P}(|z| > \tau) \approx e^{-\tau^2/2}$ for large $\tau > 0$, where $z \sim \mathcal{N}(0, 1)$.

(c) (15 points) We generalize the mutual coherence to any matrix $A = [a_1, \dots, a_p] \in \mathbb{R}^{n \times p}$, with $n \leq p$:

$$\mu(A) = \max_{1 \leq i \neq j \leq p} \frac{|a_i^\top a_j|}{\|a_i\| \|a_j\|}.$$

Show that

$$\mu(A) \geq \sqrt{\frac{p-n}{p-1}} \cdot \frac{1}{n}.$$

This is a special case of the Welch bound.

Hint: For any positive semidefinite matrix $M \in \mathbb{R}^{n \times n}$,

$$\|M\|_F^2 \geq \frac{1}{n} \left(\sum_{i=1}^n \lambda_i(M) \right)^2.$$

Problem 4. Relationship between RIC and Incoherence**[15 points]**

Let δ_k denote the restricted isometry constants for a given matrix A . Let μ be the pairwise incoherence parameter for A . Further, let $\mu_1(k)$ be the *1-coherence parameter* (or Babel function) for A , defined as

$$\mu_1(k) = \max_{S \subset [n], |S|=k} \|A_S^\top A_S\|_\infty \leq k\mu.$$

Verify that:

(a) (5 points) $\delta_2 = \mu$,

(b) (10 points) $\delta_k \leq \mu_1(k-1) \leq (k-1)\mu$.

Problem 5. Restricted Isometry Property**[25 points]**

Let $A \in \mathbb{R}^{m \times n}$ be a random matrix with i.i.d. $\mathcal{N}(0, 1/m)$ entries.

- (a) (15 points) Using concentration inequalities for sub-Gaussian matrices, prove that for any $\delta \in (0, 1)$, there exist constants $c_1, c_2 > 0$ such that the RIP constant of order s satisfies

$$\delta_s \leq \delta$$

with probability at least $1 - 2\exp(-c_2 m)$, provided that

$$m \geq c_1 \delta^{-2} s \log\left(\frac{en}{s}\right).$$

- (b) (10 points) Discuss the optimality of this bound by comparing it to the known lower bound $m \geq Cs \log(n/s)$ for any sensing matrix satisfying the RIP. What does the logarithmic factor represent in terms of the geometry of sparse vectors?

Problem 6. Covering and Packing Numbers

[20 points]

Definition 1 (ϵ -net). Let (T, d) be a metric space. Consider a set $K \subset T$ and a number $\epsilon > 0$. A subset $N \subset K$ is called an ϵ -net of K if every point in K is within distance ϵ of some point of N , i.e.,

$$\forall x \in K, \exists x_0 \in N : d(x, x_0) \leq \epsilon.$$

Definition 2 (Covering numbers). The smallest cardinality of an ϵ -net of K is called the *covering number* of K and is denoted by $N(K, d, \epsilon)$. Equivalently, $N(K, d, \epsilon)$ is the smallest number of closed balls with centers in K and radius ϵ whose union covers K .

Definition 3 (Packing numbers). A subset N of a metric space (T, d) is ϵ -separated if

$$d(x, y) > \epsilon \quad \text{for all distinct points } x, y \in N.$$

The largest possible cardinality of an ϵ -separated subset of a given set $K \subset T$ is called the *packing number* of K and is denoted by $P(K, d, \epsilon)$.

- (a) (5 point) Let N be a maximal ϵ -separated subset of K . Show that N is an ϵ -net of K .
 (b) (5 point) Prove that for any set $K \subset T$ and any $\epsilon > 0$,

$$P(K, d, 2\epsilon) \leq N(K, d, \epsilon) \leq P(K, d, \epsilon).$$

- (c) (10 points) Let (T, d) be a metric space and consider a subset $K \subset T$. Let $C(K, d, \epsilon)$ denote the smallest number of bits sufficient to specify every point $x \in K$ with accuracy ϵ in the metric d . Then

$$\log_2 N(K, d, \epsilon) \leq C(K, d, \epsilon) \leq \lceil \log_2 N(K, d, \frac{\epsilon}{2}) \rceil.$$

Problem 7. Mutual Incoherence (Bonus)

[15 points]

Let $X \in \mathbb{R}^{n \times p}$ be a design matrix with ℓ_2 -normalized columns and mutual incoherence

$$\mu(X) = \max_{j \neq k} |\langle X_j, X_k \rangle|.$$

The Lasso estimator is defined as

$$\hat{\theta} \in \arg \min_{\theta \in \mathbb{R}^p} \left\{ \frac{1}{2n} \|y - X\theta\|_2^2 + \lambda_n \|\theta\|_1 \right\}.$$

Prove that for the Lasso to achieve exact support recovery (i.e., $\text{supp}(\hat{\theta}) = \text{supp}(\theta^*)$) with high probability in the noiseless case, a necessary condition is given by

$$\min_{j \in S} |\theta_j^*| > \frac{\lambda_n}{1 - \mu(X)(s - 1)},$$

where S is the true support of size s .